A Comparative Study of Machine Learning Algorithms for Heart Disease Prediction

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***Abstract*—Heart disease is a major health problem, and early prediction can support timely identification of patients at higher risk. This study reimplements a machine learning-based approach for heart disease prediction using the UCI Heart Disease dataset, which contains 303 patient records with 13 input features. During preprocessing, missing values were handled using mode imputation, and the target variable was converted into a binary classification representing the presence or absence of heart disease. Three machine learning algorithms, namely K-Nearest Neighbors (KNN), Naive Bayes, and Decision Tree, were implemented and compared. Hyperparameter tuning was performed using GridSearchCV, and model performance was evaluated using 10-fold cross-validation. Among the evaluated models, Naive Bayes achieved the best overall performance, with an accuracy of 83.83%, precision of 84.62%, recall of 79.14%, and F1-score of 81.78%. KNN achieved an accuracy of 82.84%, while Decision Tree achieved 79.87%. The mean cross-validation accuracy was 83.78% for Naive Bayes, 82.80% for KNN, and 79.80% for Decision Tree. The results indicate that Naive Bayes provided the strongest overall classification performance for the selected dataset and experimental setup. This study demonstrates the effectiveness of machine learning-based classification for heart disease prediction while reimplementing the methodology of the selected research work.**

***Keywords*—Heart Disease Prediction, Machine Learning, KNN, Naive Bayes, Decision Tree, UCI Heart Disease Dataset**

I. Introduction

Heart disease is one of the major health problems worldwide and remains an important cause of death. Its development is associated with several risk factors, including high blood pressure, high cholesterol, diabetes, smoking, age, and other lifestyle-related factors [1]. Early identification of heart disease can support timely medical decisions and improve patient care. However, accurate diagnosis can be challenging, particularly in situations where access to specialized medical resources is limited. Machine Learning has therefore gained attention as a useful approach for analyzing patient data and supporting heart disease prediction.

Machine Learning classification algorithms can identify patterns in patient information and use these patterns to predict whether a patient is likely to have heart disease [2]. However, the performance of a model can vary depending on the dataset, preprocessing techniques, feature handling, and algorithm used. For this reason, comparing different classification algorithms under the same experimental conditions is useful for understanding their strengths and limitations.

This study reimplements the general heart disease prediction approach presented in a previously published study. The original research investigates the use of Machine Learning techniques for predicting heart disease and evaluates the performance of different classification methods. Based on this general methodology, this project focuses on three Machine Learning algorithms covered in the course: K-Nearest Neighbors (KNN), Naive Bayes, and Decision Tree. The aim is not to reproduce the original study exactly, but to implement and evaluate the selected algorithms using a consistent and reproducible experimental setup.

The UCI Heart Disease dataset is used for the implementation. It contains 303 patient records with 13 input features related to patient health and cardiovascular risk factors. Before model evaluation, the dataset is preprocessed to make it suitable for classification. Missing values are handled using mode imputation, and the original target variable is converted into a binary target representing the absence or presence of heart disease. Feature analysis is also performed to better understand the dataset and the relationship between the input features and the target variable.

To evaluate the models, **10-fold cross-validation** is used. Hyperparameter tuning is performed using *GridSearchCV* with 10-fold cross-validation to identify suitable parameter configurations for each algorithm. The selected hyperparameters are then used for the final model evaluation using the same 10-fold cross-validation framework. The performance of the three algorithms is measured using accuracy, precision, recall, F1-score, and confusion matrices. Using multiple evaluation measures provides a more complete view of model performance rather than relying only on accuracy.

The main objective of this study is to implement a Machine Learning-based heart disease prediction workflow and compare KNN, Naive Bayes, and Decision Tree using the same dataset and experimental conditions. The comparison helps determine which of the three selected algorithms provides better predictive performance for the given heart disease dataset. The study also demonstrates how data preprocessing, feature analysis, hyper-parameter tuning, cross-validation, and performance evaluation

can be combined to develop a simple and reproducible Machine Learning-based prediction system.

II. Related Work

The selected research work by Ansari et al. presented a Machine Learning-based approach for heart disease prediction using different classification techniques [3].

The study focused on evaluating Machine Learning techniques for classifying patients based on the presence or absence of heart disease. The authors considered several steps in their approach, including data preprocessing, feature analysis, data splitting, K-fold cross-validation, application of Machine Learning algorithms, and performance evaluation [4].

The research used a heart disease dataset containing patient information and a target value representing the presence or absence of heart disease. The study applied different Machine Learning classification techniques, including K-Nearest Neighbors (KNN), Naive Bayes, Decision Tree, Logistic Regression, and Support Vector Machine (SVM) [5]. The models were evaluated using performance measures such as accuracy and other classification metrics.

An important part of the proposed approach was the analysis and selection of relevant features before model training. The study also used K-fold cross-validation to reduce the effect of a particular data split and to evaluate the models more consistently. These steps provide a basis for comparing the performance of different Machine Learning algorithms under similar experimental conditions.

This research work was selected as the basis for the present project because its overall workflow is suitable for a Machine Learning course project. In this reimplementation, the general workflow of the research is followed while using three algorithms covered in the course: KNN, Naive Bayes, and Decision Tree. The models were implemented using the selected UCI Heart Disease dataset and evaluated using 10-fold cross-validation after hyperparameter tuning with *GridSearchCV*. Their performance was compared using accuracy, precision, recall, and F1-score.

III. Dataset and Methodology

A. Dataset

The UCI Heart Disease dataset was used in this study. The original UCI Heart Disease database contains 76 attributes, while the Cleveland dataset used in this study contains 14 attributes, including 13 input attributes and one target attribute. The dataset contains 303 patient records.

The 14 attributes used in this study are presented in Table I.

The 13 input attributes used for model development are *age*, *sex*, *cp*, *trestbps*, *chol*, *fbs*, *restecg*, *thalach*, *exang*, *oldpeak*, *slope*, *ca*, and *thal*.

The original *num* attribute contains five possible values ranging from 0 to 4. A value of 0 represents the absence of heart disease, whereas values from 1 to 4 indicate the presence of heart disease. In this study, the original target was converted into a binary classification problem using the following rule:

$$Target = \begin{cases} 0, & \text{if } Num = 0 \\ 1, & \text{if } Num > 0 \end{cases} \tag{1}$$

After target conversion, the *num* attribute was renamed to *target*. Therefore, the final dataset consists of 303 records, 13 input attributes, and one binary target attribute.

The final attributes used in the study are: *age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, target*.

Thus, this study formulates heart disease prediction as a binary classification problem, where 0 represents the absence of heart disease and 1 represents its presence.

The original dataset is publicly available from the UCI Machine Learning Repository at:

https://archive.ics.uci.edu/dataset/45/heart+disease

B. Data Preprocessing

Data preprocessing was performed before applying the machine learning algorithms. The dataset was first checked for missing values and duplicate records. The dataset contained 303 records, and no duplicate rows were found.

Two features contained missing values. The *ca* feature had 4 missing values, while the *thal* feature had 2 missing values. Since the dataset contains only 303 records, removing these rows could reduce the amount of available data. Therefore, the missing values were imputed instead of removing the corresponding records.

Mode imputation was used to fill the missing values. The mode is the value that appears most frequently in a feature [6]. It was selected because *ca* and *thal* represent categorical or discrete values in this dataset. The mode value for *ca* was 0, while the mode value for *thal* was 3.

After imputation, all 303 records contained valid values for all 13 input features. This ensured that the complete dataset could be used for subsequent feature analysis, hyperparameter tuning, and machine learning model evaluation.

C. Feature Analysis

After handling the missing values, the features were analyzed before applying the machine learning models. The final dataset contained 13 input features. These features were divided into numerical and categorical features based on their values and meaning in the dataset [7].

The numerical features were *age*, *trestbps*, *chol*, *thalach*, and *oldpeak*. The categorical or discrete features were *sex*, *cp*, *fbs*, *restecg*, *exang*, *slope*, *ca*, and *thal*.

The distribution of the features was examined to better understand the characteristics of the dataset before applying the machine learning models. The numerical features were

TABLE I
FEATURES OF THE UCI HEART DISEASE DATASET

S.No.	Attributes	Description
1	Age	Age of the patient in years.
2	Sex	Gender of the patient.
3	CP	Type of chest pain.
4	Trestbps	Resting blood pressure in mm Hg.
5	Chol	Serum cholesterol level in mg/dl.
6	FBS	Fasting blood sugar level.
7	Restecg	Resting electrocardiogram result.
8	Thalach	Maximum heart rate achieved.
9	Exang	Exercise-induced angina.
10	Oldpeak	ST depression caused by exercise.
11	Slope	Slope of the peak exercise ST segment.
12	Ca	Number of major vessels detected.
13	Thal	Thalassemia result.
14	Num	Heart disease diagnosis.

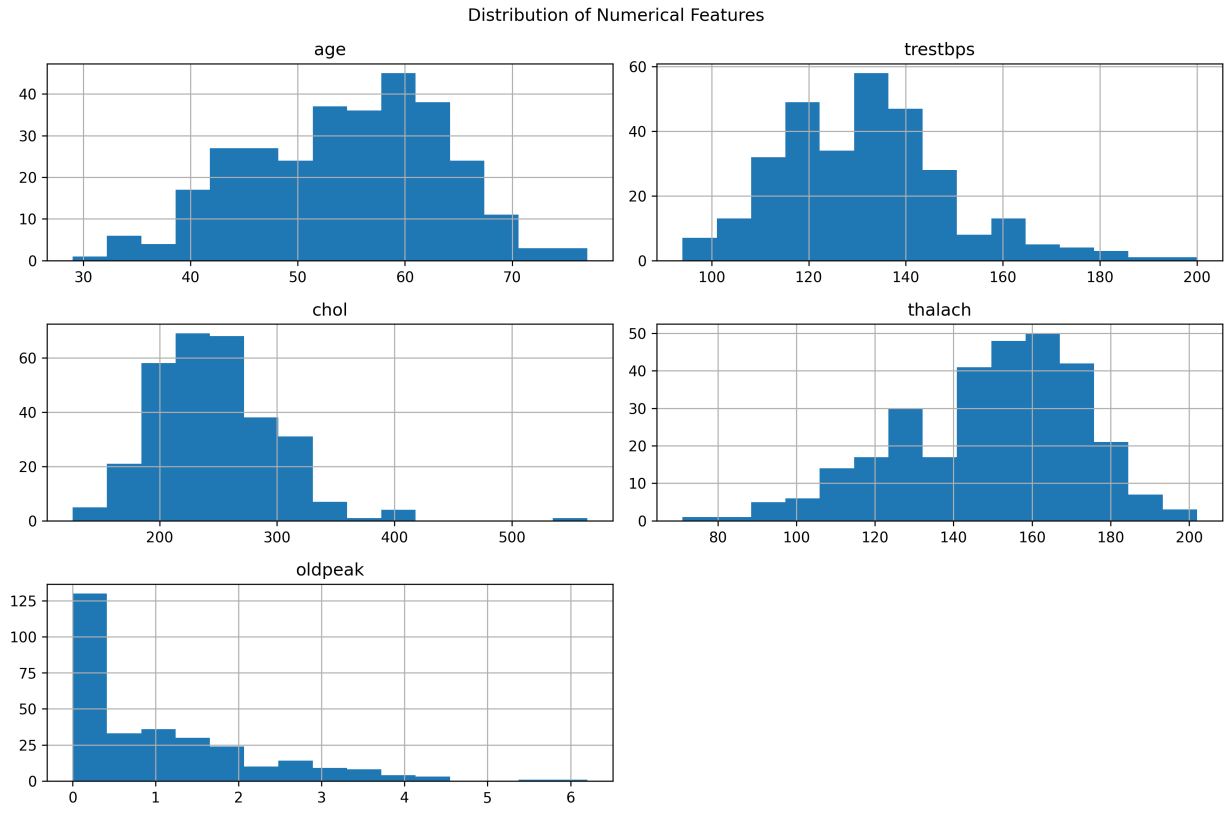


Fig. 1. Distribution of numerical features in the dataset.

analyzed using their minimum, maximum, mean, and standard deviation values. The categorical features were also examined to identify the different values and their distributions within each feature.

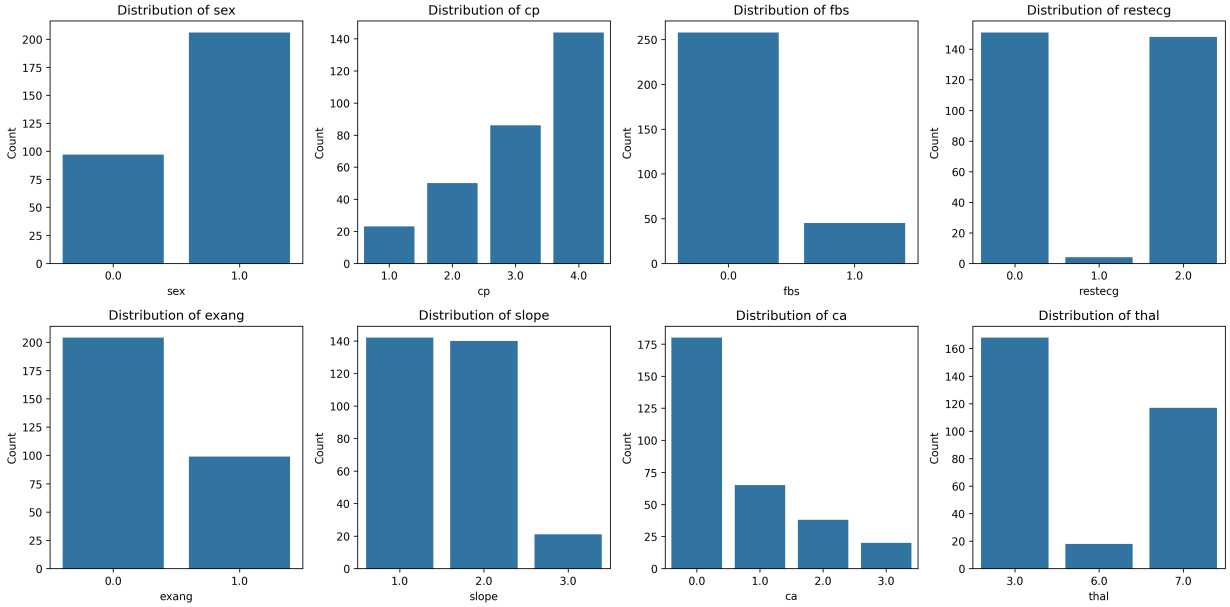


Fig. 2. Distribution of categorical features in the dataset.

This feature analysis provided an overview of the dataset and helped verify the characteristics of the input variables before hyperparameter tuning and 10-fold cross-validation.

D. Correlation Analysis

Correlation analysis was performed to study the relationship between the input features and the target variable. A correlation matrix was used to visualize the relationships among the features and the target variable.

The correlation analysis showed that *thal*, *ca*, *exang*, *oldpeak*, and *cp* had relatively higher absolute correlations with the target. The *thalach* feature showed a relatively strong negative correlation with the target. In comparison, *fbs* and *chol* showed relatively low correlations with the target.

The features were also ranked based on their absolute correlation with the target variable to understand which features had stronger relationships with the prediction target.

Although some features showed stronger correlations with the target than others, all 13 input features were retained for the model evaluation. Therefore, correlation analysis was used to understand the relationships between the input features and the target variable rather than to remove features based only on their correlation values.

The retained features were subsequently used for hyperparameter tuning and 10-fold cross-validation of the three machine learning models.

E. 10-Fold Cross-Validation

To evaluate the machine learning models reliably, 10-fold cross-validation was used. In this method, the complete preprocessed dataset was divided into 10 folds. In each iteration, one fold was used as the validation set, while the remaining nine folds were used for model fitting. This process was repeated until every fold had been used once as the validation set.

Since the dataset contains 303 records, the validation folds contained either 30 or 31 records. The first three folds contained 31 validation records, while the remaining seven folds contained 30 validation records. The corresponding model fitting sets contained 272 or 273 records, depending on the iteration.

The 10-fold cross-validation process can be represented as:

$$D = D_1 \cup D_2 \cup \cdots \cup D_{10} \tag{2}$$

where D represents the complete dataset and each D_i represents one fold. During each iteration, one fold was used for validation and the other nine folds were used for model fitting.

The same 10-fold cross-validation procedure was applied to KNN, Naive Bayes, and Decision Tree to ensure a fair comparison under the same experimental conditions. The cross-validation procedure was also used within *GridSearchCV* for hyperparameter tuning. The best hyperparameter configuration for each model was selected based on its cross-validation performance.

The accuracy obtained from each fold was recorded, and the mean accuracy was calculated to represent the overall cross-validation performance of each model. The standard deviation was also calculated to measure the variation in accuracy across the different folds.

The mean cross-validation accuracy was calculated as:

$$Mean\ Accuracy = \frac{1}{k}\sum_{i=1}^{k} Accuracy_i \tag{3}$$

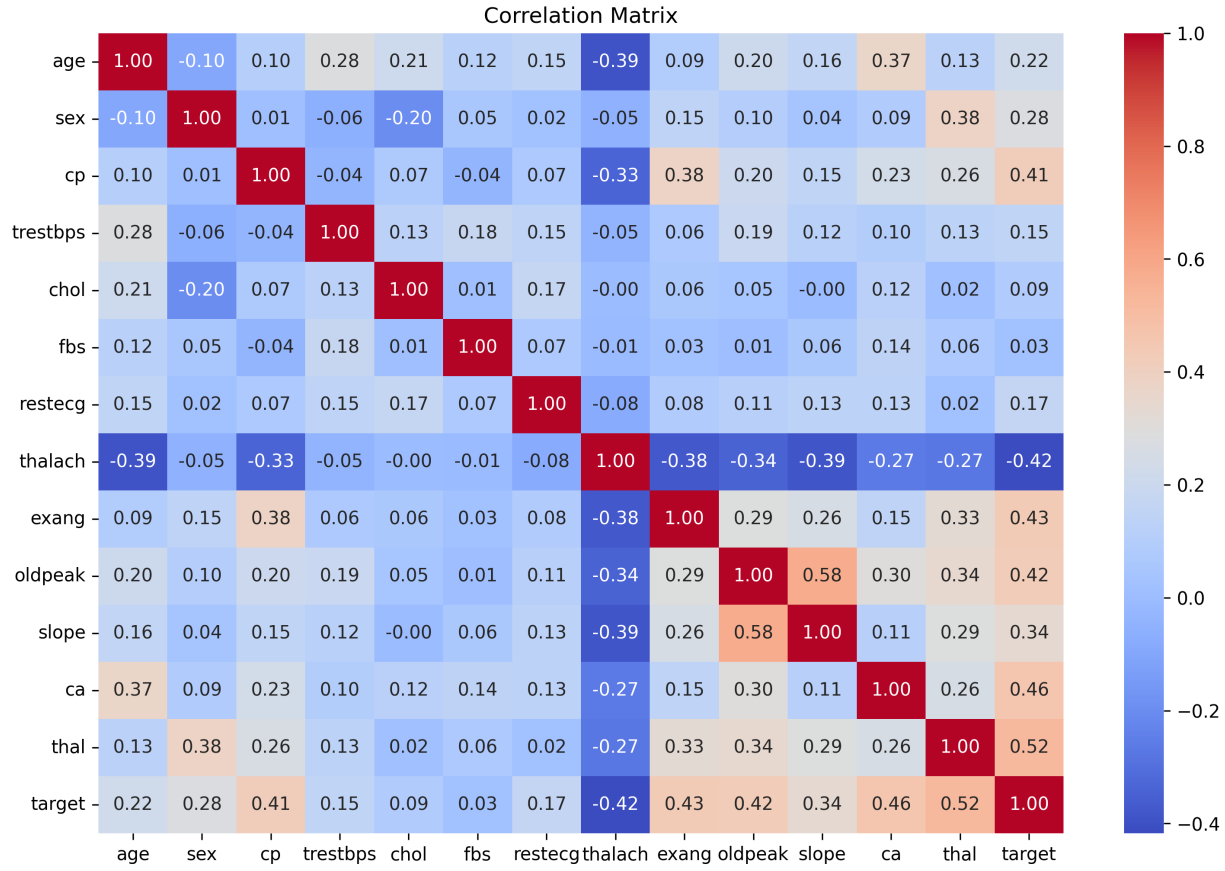


Fig. 3. Correlation matrix of the input features and target.

where $k = 10$ represents the number of folds and $Accuracy_i$ represents the accuracy obtained from the i-th fold.

The standard deviation was calculated to measure the variation in model accuracy across the ten validation folds. A lower standard deviation indicates more consistent performance across the folds.

F. Machine Learning Algorithms

Three supervised machine learning classification algorithms were used in this study: K-Nearest Neighbors (KNN), Naive Bayes, and Decision Tree. These algorithms were selected because they are covered in the course and can be applied to the binary heart disease classification problem. Hyperparameter tuning was performed using *GridSearchCV*, and the models were evaluated using 10-fold cross-validation.

1) K-Nearest Neighbors: K-Nearest Neighbors (KNN) is a supervised classification algorithm that predicts the class of a new data sample by considering its nearest samples in the dataset. The value of K determines how many neighboring samples are considered for the prediction.

For a new patient record, KNN calculates the distance between the new record and the available samples. In this study, **Euclidean distance** was selected as the distance metric based on the hyperparameter tuning results. The Euclidean distance between two samples can be calculated as:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{4}$$

where x_i and y_i represent the values of the i-th feature in the two samples, and n represents the total number of features.

After calculating the distances, the $K = 13$ nearest samples are selected. In this study, **distance-based weights** are used, meaning that closer neighbors have a greater influence on the prediction. The weighted votes of the nearest neighbors are then used to determine the predicted class.

2) Naive Bayes: Naive Bayes is a supervised classification algorithm based on Bayes' theorem. It calculates the probability of each possible class based on the available input features and assigns the sample to the class with the highest posterior probability.

The basic form of Bayes' theorem is given by:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)} \tag{5}$$

where C represents a class and X represents the input features. $P(C|X)$ is the posterior probability of the class given

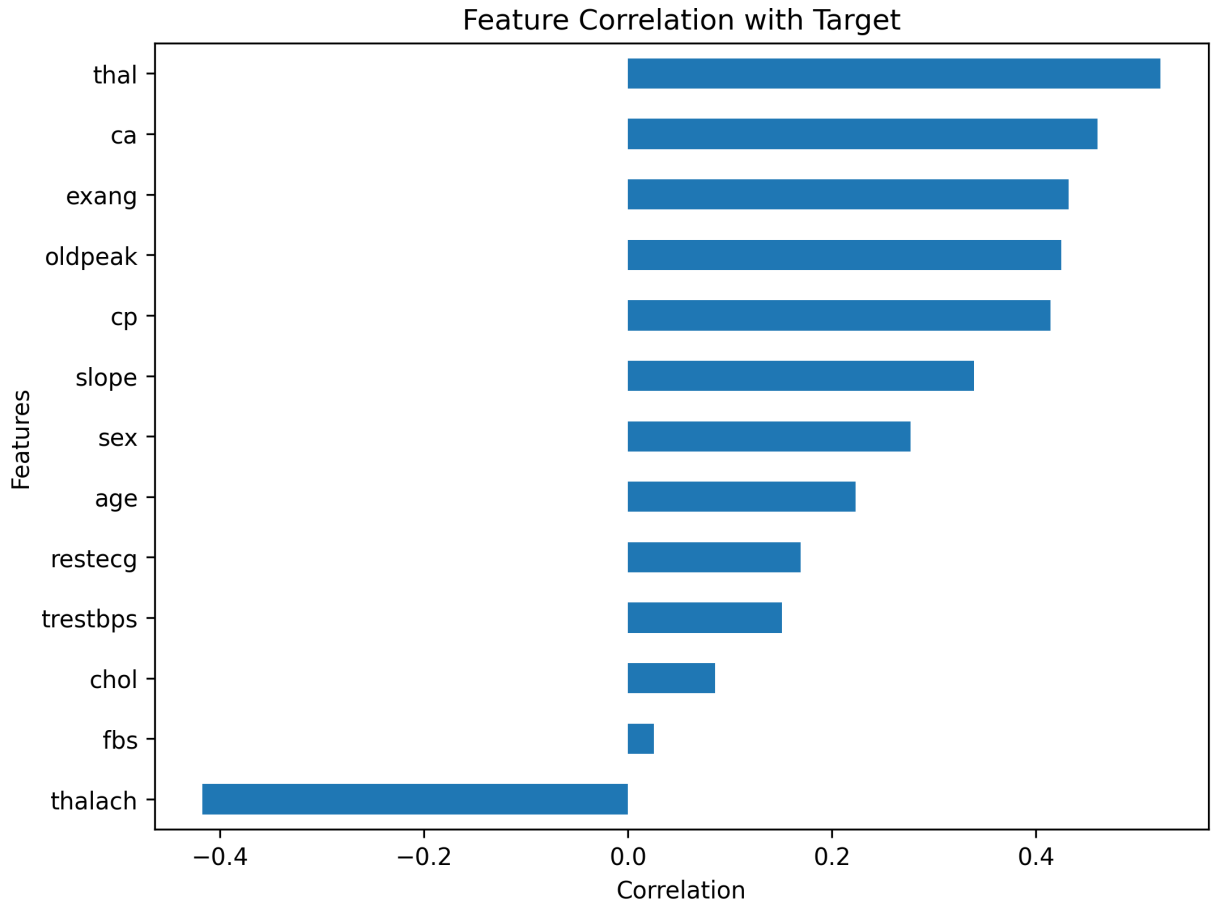


Fig. 4. Correlation of input features with the target variable.

the input features, $P(X|C)$ is the likelihood, $P(C)$ is the prior probability of the class, and $P(X)$ is the probability of the input features.

The method assumes that the input features are conditionally independent given the class. Although this assumption may not always be completely true for medical data, Naive Bayes is simple, computationally efficient, and suitable for binary classification tasks.

3) Decision Tree: Decision Tree is a supervised classification algorithm that makes predictions using a tree-like structure. The tree contains decision nodes, branches, and leaf nodes. Each decision node checks a feature and divides the data into different groups based on a selected splitting criterion.

The process continues recursively until the model reaches a leaf node, which represents the predicted class. Decision Trees are easy to understand because their decisions can be represented as a series of simple rules.

IV. Experimental Setup

The experiments were carried out using Python and standard machine learning libraries in a Jupyter Notebook environment. The same preprocessed dataset was used for all three machine learning models to provide a fair and consistent comparison.

The dataset contains 303 records and 13 input features. The target variable was converted into two classes, where 0 represents the absence of heart disease and 1 represents the presence of heart disease.

A. Hyperparameter Tuning

Hyperparameter tuning was performed using *GridSearchCV* to identify suitable parameter configurations for each machine learning model. A 10-fold cross-validation strategy was used during the tuning process, and the best hyperparameter combination was selected based on cross-validation performance.

The selected hyperparameters were then used to evaluate each model using the same 10-fold cross-validation procedure. This ensured that all three models were evaluated under consistent experimental conditions.

B. KNN Configuration

The K-Nearest Neighbors (KNN) classifier was configured using the best parameters obtained from *GridSearchCV*. The selected configuration used $K = 13$, Euclidean distance, and distance-based weights.

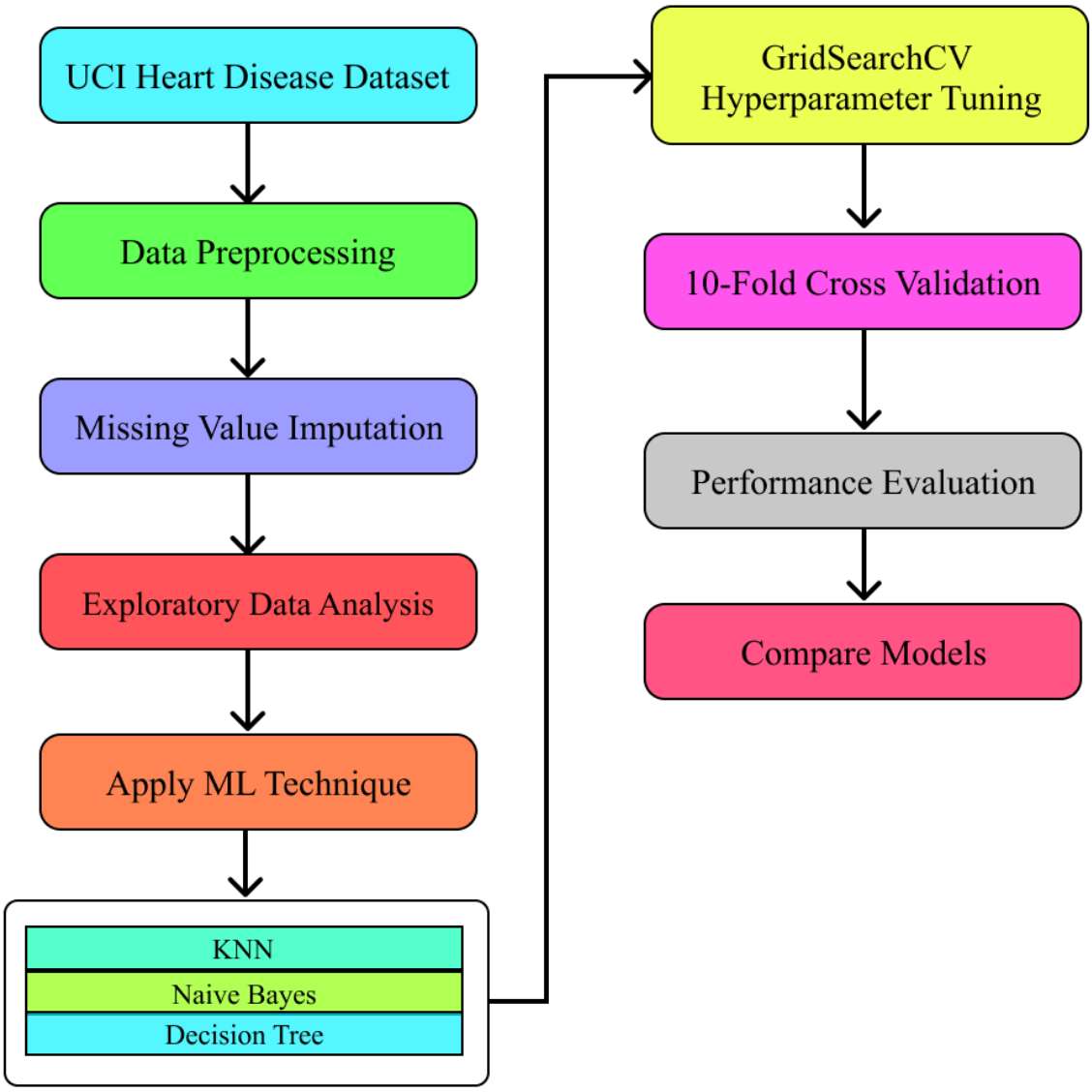


Fig. 5. Methodology Architecture for Heart Disease Prediction

The tuned KNN model was evaluated using 10-fold cross-validation. The accuracy obtained from each fold was recorded, and the mean accuracy and standard deviation were calculated to summarize its cross-validation performance.

C. Naive Bayes Configuration

The Naive Bayes classifier was used as a probabilistic classification model. The *var_smoothing* parameter was tuned using *GridSearchCV* with 10-fold cross-validation. The best value obtained was 1×10^{-11}.

The tuned Naive Bayes model was evaluated using the same 10-fold cross-validation procedure as KNN. Its performance was then compared with KNN and Decision Tree using the selected evaluation metrics.

D. Decision Tree Configuration

A Decision Tree classifier was implemented for the binary classification task. Its main hyperparameters were tuned using *GridSearchCV* with 10-fold cross-validation.

The best configuration used the Gini criterion, a maximum depth of 3, a minimum leaf size of 10 samples, and a minimum split size of 5 samples.

The tuned Decision Tree was evaluated using the same 10-fold cross-validation procedure used for KNN and Naive Bayes. This ensured that all three models were evaluated under the same experimental conditions.

E. Cross-Validation

Ten-fold cross-validation was used to evaluate the models across multiple training and validation folds. In each iteration,

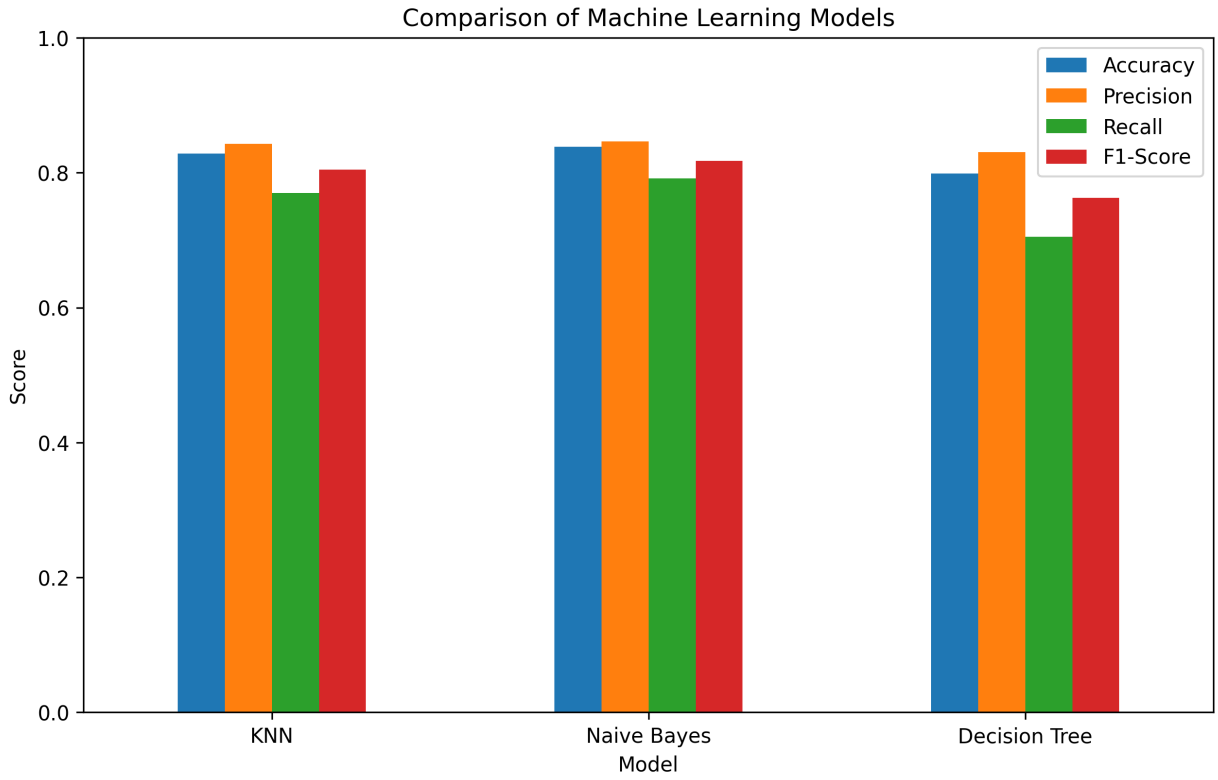


Fig. 6. Comparison of KNN, Naive Bayes, and Decision Tree using 10-fold cross-validation.

nine folds were used for model fitting and one fold was used for validation. This process was repeated until every fold had been used once as the validation fold.

The accuracy from each fold was recorded for all three models. The mean accuracy and standard deviation were then calculated to summarize the cross-validation performance.

The mean accuracy for a model was calculated as:

$$\bar{A} = \frac{1}{k} \sum_{i=1}^{k} A_i \tag{6}$$

where $k = 10$ represents the number of folds and A_i represents the accuracy obtained from the i-th fold.

The standard deviation was also calculated to measure the variation in accuracy across the ten folds. A lower standard deviation indicates more consistent performance across the different validation folds.

F. Performance Metrics

The models were evaluated using accuracy, precision, recall, and F1-score. A confusion matrix was also generated for each model to provide a detailed view of correct and incorrect classification results.

Accuracy measures the overall proportion of correctly classified samples. Precision measures the proportion of samples predicted as positive that are actually positive. Recall measures the proportion of actual positive samples that are correctly identified. F1-score combines precision and recall into a single measure.

The metrics were calculated using the following equations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10}$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative.

V. Results and Discussion

The performance of KNN, Naive Bayes, and Decision Tree was evaluated using 10-fold cross-validation. The tuned hyperparameters obtained through *GridSearchCV* were used for the evaluation. Accuracy, precision, recall, F1-score, and confusion matrices were used to compare the performance of the three models.

A. 10-Fold Cross-Validation Results

The 10-fold cross-validation method was used to evaluate the performance of the three machine learning models. The complete preprocessed dataset was divided into ten folds. In each iteration, one fold was used for validation and the remaining nine folds were used for model fitting. The process was repeated until every fold had been used once for validation.

The selected hyperparameters obtained through *GridSearchCV* were used during the evaluation. The accuracy obtained from each fold was recorded, and the mean accuracy and standard deviation were calculated. Accuracy, precision, recall, and F1-score were also calculated to compare the overall performance of the models. Table II presents the results obtained from the 10-fold cross-validation.

TABLE II
10-FOLD CROSS-VALIDATION RESULTS

Model	Accuracy	Precision	Recall	F1-Score
KNN	82.84%	84.25%	76.98%	80.45%
Naive Bayes	**83.83%**	**84.62%**	**79.14%**	**81.78%**
Decision Tree	79.87%	83.05%	70.50%	76.26%

Figure 6 provides a visual comparison of the performance of the three machine learning models.

- **Naive Bayes:** Achieved the highest accuracy of **83.83%**, precision of **84.62%**, recall of **79.14%**, and F1-score of **81.78%**. It provided the best overall performance among the three models.
- **KNN:** Achieved an accuracy of **82.84%**, precision of **84.25%**, recall of **76.98%**, and F1-score of **80.45%**. It achieved the second-highest overall performance among the three models.
- **Decision Tree:** Achieved an accuracy of **79.87%**, precision of **83.05%**, recall of **70.50%**, and F1-score of **76.26%**. It showed lower performance than both KNN and Naive Bayes across the main evaluation metrics.

Overall, Naive Bayes provided the best overall performance in the 10-fold cross-validation evaluation, followed by KNN and Decision Tree.

1) KNN Confusion Matrix: The confusion matrix for KNN obtained during the cross-validation evaluation is shown in Figure 7.

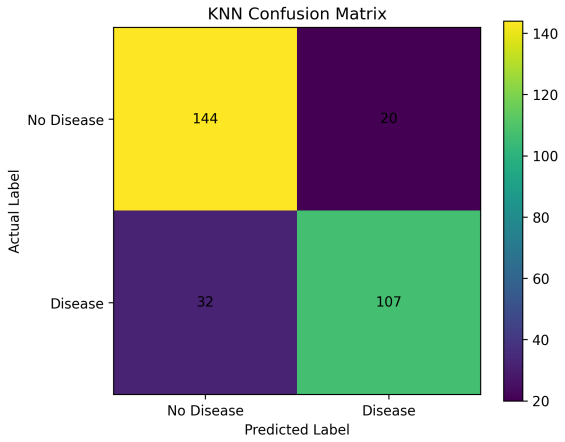


Fig. 7. KNN confusion matrix for the 10-fold cross-validation evaluation.

2) Naive Bayes Confusion Matrix: The confusion matrix for Naive Bayes obtained during the cross-validation evaluation is shown in Figure 8.

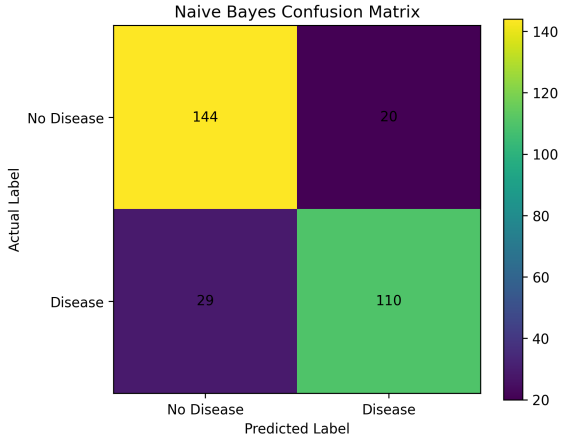


Fig. 8. Naive Bayes confusion matrix for the 10-fold cross-validation evaluation.

3) Decision Tree Confusion Matrix: The confusion matrix for Decision Tree obtained during the cross-validation evaluation is shown in Figure 9.

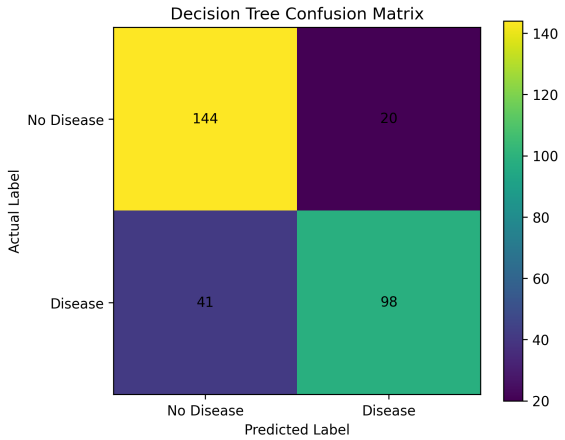


Fig. 9. Decision Tree confusion matrix for the 10-fold cross-validation evaluation.

VI. Conclusion

This study implemented and compared three machine learning algorithms, namely K-Nearest Neighbors (KNN), Naive Bayes, and Decision Tree, for heart disease prediction. The experiments were conducted using the UCI Heart Disease Dataset, consisting of 303 patient records and 13 input features. Missing values were handled using mode imputation, and the target variable was converted into a binary classification representing the absence or presence of heart disease. Hyperparameter tuning was performed using *GridSearchCV* to identify suitable parameter configurations for each model.

The tuned models were evaluated using 10-fold cross-validation. Among the three models, Naive Bayes achieved the best overall performance, with an accuracy of 83.83%, precision of 84.62%, recall of 79.14%, and F1-score of 81.78%. KNN achieved an accuracy of 82.84%, precision of 84.25%, recall of 76.98%, and F1-score of 80.45%. Decision Tree achieved an accuracy of 79.87%, precision of 83.05%, recall of 70.50%, and F1-score of 76.26%.

The fold-wise cross-validation results showed that KNN achieved a mean accuracy of 82.80% with a standard deviation of 6.16%. Naive Bayes achieved the highest mean accuracy of 83.78% with a standard deviation of 6.10%, while Decision Tree achieved a mean accuracy of 79.80% with a standard deviation of 7.92%. The relatively lower standard deviation of Naive Bayes compared with Decision Tree indicates more consistent performance across the validation folds.

The best hyperparameter configuration for KNN was $K = 13$ with Euclidean distance and distance-based weights. For Naive Bayes, the best *var_smoothing* value was 1×10^{-11}. For Decision Tree, the best configuration used the Gini criterion, a maximum depth of 3, a minimum leaf size of 10 samples, and a minimum split size of 5 samples.

The study followed the major components of the methodology described in the selected research paper, including data preprocessing, feature analysis, machine learning classification, hyperparameter tuning, and 10-fold cross-validation. However, the results of this reimplementation are not expected to be exactly the same as those reported in the original paper because differences in preprocessing, feature handling, implementation details, hyperparameter settings, and experimental conditions can affect the final results.

Overall, the experiment demonstrates that traditional machine learning algorithms can provide useful performance for heart disease prediction. Among the evaluated models, Naive Bayes provided the best overall performance based on accuracy, precision, recall, and F1-score. Therefore, Naive Bayes was identified as the best-performing model for the selected dataset and experimental setup.

TABLE III
Final Results of 10-Fold Cross-Validation

Model	Accuracy	Precision	Recall	F1-Score
KNN	82.84%	84.25%	76.98%	80.45%
Naive Bayes	**83.83%**	**84.62%**	**79.14%**	**81.78%**
Decision Tree	79.87%	83.05%	70.50%	76.26%

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